

Principles of Cost Accounting (1)

Questions, correct answers, and reasoning based on the provided lectures

All answers below are sourced 100% from the lectures you provided (Chapters 1-2, Chapter 4, and the disposition of under- or over-applied MOH). No external sources were needed.

Question 1

Which of the following is normally assigned to a cost object through cost tracing?

- | | |
|----|---------------------------------|
| a. | D. Manufacturing overhead only |
| b. | A. Indirect cost |
| c. | B. Direct cost [CORRECT] |
| d. | C. Common fixed cost |

Correct answer: c. B. Direct cost

Reasoning

- The lecture defines the two ways of assigning cost to a cost object. Cost tracing is used for direct costs, and cost allocation is used for indirect costs.
- A direct cost is related to a certain cost object and can be traced to it in an economically feasible way. An indirect cost cannot be traced to any cost object, so it is allocated using a cost allocation base.
- Therefore, the item assigned through cost tracing is the direct cost.

Lecture reference

- "Cost tracing is the assignment of direct cost to a cost object."
- "Cost allocation is the assignment of indirect cost to a cost object."
- "We can't trace indirect cost directly to any cost object but we allocate this cost to different cost objects using cost allocation base."

Why the other options are wrong

- A. Indirect cost: assigned by cost allocation, not tracing.
- C. Common fixed cost: a common cost is shared by many cost objects, so it is indirect and is allocated.
- D. Manufacturing overhead only: the lecture says MOH is an indirect cost related to all job orders, so it must be accumulated in a cost pool and allocated.

Verification: answer is 100% sourced from the lecture.

Question 2

Within the relevant range, when production volume increases:

- a. B. Variable cost per unit decreases
- b. C. Total variable cost remains constant
- c. A. Total fixed cost increases proportionately
- d. **D. Fixed cost per unit normally decreases [CORRECT]**

Correct answer: d. D. Fixed cost per unit normally decreases

Reasoning

- Total fixed cost stays the same when volume changes, so spreading the same total over more units makes the fixed cost per unit fall.
- The lecture's own example shows this: with total fixed cost of \$10,000, average fixed cost falls from 100 to 50 to 33.33 to 25 as volume rises from 100 to 400 units.

Lecture reference

- "Fixed cost per unit is changing (decreasing) when volume of output or level of activity is increasing."
- "Average fixed cost decreases as units increase."

Why the other options are wrong

- A. Total fixed cost does not increase: it "doesn't change regardless of the change in level of activity or volume of production" (within its relevant range).
- B. Variable cost per unit does not decrease: "Variable cost per unit is constant."
- C. Total variable cost does not stay constant: it "changes in proportion to the change in volume of output."

Verification: answer is 100% sourced from the lecture.

Question 3

Which equation represents a mixed cost?

- a. $Y = A$
- b. $Y = A * X$
- c. $Y = BX$
- d. **$Y = A + BX$ [CORRECT]**

Correct answer: d. $Y = A + BX$

Reasoning

- A mixed cost includes both a variable part and a fixed part. In the lecture's equation, A is the fixed cost and BX is the variable cost (B = variable cost per unit, X = cost driver).
- Only $Y = A + BX$ contains both parts.

Lecture reference

- "Mixed cost includes both variable cost & fixed cost."
- "Total cost equation: $Y = A + BX$. Where: Y: total cost. A: fixed cost. B: variable cost per unit. X: cost driver."

Why the other options are wrong

- a. $Y = A$ is the lecture's fixed cost equation.
- c. $Y = BX$ is the lecture's variable cost equation.
- b. $Y = A * X$ does not appear in the lecture as a cost equation, and it has no separate fixed component.

Verification: answer is 100% sourced from the lecture.

Question 4

Maintenance cost is \$14,000 at 2,000 machine-hours and \$26,000 at 5,000 machine-hours. Using the high-low method, the variable cost per machine-hour is:

- a. B. \$6
- b. D. \$12
- c. C. \$8
- d. A. \$4 [CORRECT]

Correct answer: d. A. \$4 per machine-hour

Reasoning

- The lecture finds B (average variable cost per unit) with the maximum and minimum levels: $B = (\text{cost at maximum level} - \text{cost at minimum level}) / (\text{maximum level} - \text{minimum level})$.
- Maximum level = 5,000 hours (\$26,000). Minimum level = 2,000 hours (\$14,000).
- $B = (26,000 - 14,000) / (5,000 - 2,000) = 12,000 / 3,000 = \4 per machine-hour.
- Check, using the lecture's substitution step to find fixed cost: $26,000 = A + 4 \times 5,000$, so $A = \$6,000$. At the minimum level, $6,000 + 4 \times 2,000 = 14,000$, which matches the given cost.

Lecture reference

- "B = The cost of maximum level - the cost of minimum level / maximum Level - minimum Level" (also written B = change in cost / change in level).
- Lecture example: " $B = \$10,000 - \$6,000 / 500 \text{ h} - 100 \text{ h} = \$4,000 / 400 \text{ h} = \10 per hour."
- Note: the lecture does not use the name "high-low method" but describes the identical procedure as "the method of maximum volume (level) and minimum volume (level)."

Why the other options are wrong

- \$6, \$8 and \$12 do not result from $12,000 / 3,000$. (\$12 would only be the cost difference in thousands, without dividing by the change in hours.)

Verification: answer is 100% sourced from the lecture.

Question 5

Manufacturing overhead includes direct materials and direct labor.

- a. True
- b. False [CORRECT]

Correct answer: False

Reasoning

- The lecture defines manufacturing overhead as every manufacturing cost except direct materials and direct labor. It is made up of indirect materials, indirect labor, and other manufacturing overhead.
- In the lecture's formulas, direct materials, direct labor and manufacturing overhead are three separate components: Manufacturing costs = direct material + direct labor + manufacturing overhead. So MOH cannot include DM and DL.

Lecture reference

- "Any manufacturing cost except direct materials or direct labor are classified as manufacturing overhead."
- "Manufacturing overhead is divided into three categories which are: a) Indirect materials. b) Indirect labor. c) Other manufacturing overhead."
- "Manufacturing costs = direct material + direct labor + manufacturing overhead."

Verification: answer is 100% sourced from the lecture.

Answer Summary

Question	Correct answer
Q1	c. B. Direct cost
Q2	d. D. Fixed cost per unit normally decreases
Q3	d. $Y = A + BX$
Q4	d. A. \$4 per machine-hour
Q5	False